

# Towards an Encyclopedia of Programming Languages Anchored in Software Heritage & CodeCommons

Every Programming Language, with Its Real Programs

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Software Heritage



CodeCommons

PL Catalog: [blog.mathieuacher.com/PL-ultimate-llm](http://blog.mathieuacher.com/PL-ultimate-llm) · [github.com/acherm/PL-ultimate-llm](https://github.com/acherm/PL-ultimate-llm)

### Programming languages + example programs

Browse the repository without dumping everything on one page.

14,242  
PL pages

3,842  
in-rop (LLM-curated)

3,846  
LLM programs

1,242 (9.9%)  
PLs with ≥1 ext claim  
(of 12,608 in taxonomy)

192  
PLs with SWH samples

292  
SWH samples

2026-06-12  
last build (UTC)

14,242 language pages, cross-referenced with 7 sources + LLM contributions

.m

12 claimants8 primary3 secondaryshared primary (8)

#### SWH popularity

From the SWH-MSR-ARV dataset (Desmazières, Di Cosmo, Lorentz, MSR 2025; file nb\_extensions\_alphanum.csv) — one row per (ext, year) in the full SWH archive. Case-aggregated. [citation](#).

46.5M  
total occurrences

7.8M  
since 2019 (16.7%)

1970–2023  
years active

Case variants in archive: .m (46.5M), .M (26.1K) — aggregated above.

#### Attribution status

This extension is attributed to [GNU Octave](#); [GNU Octave](#); [m4](#); [Mercury](#); [Mercury](#); [Monkey C](#); [Objective-C](#); [Objective-C](#); [Objective-C](#); [octave](#); [Win32 Message File](#); [Wolfram Language](#) as primary by 13 authoritative claims.

Authoritative sources: [Linguist](#) [Pygments](#) [Wikidata](#) [Q169478](#) [Wikipedia](#) · secondary

[Disagree or have a correction? Open a labelling issue.](#)

One page per extension: claimants, SWH popularity, attribution status

#### Real programs from Software Heritage

6 samples mined from derived\_datasets/<date>/contents/\*.parquet, byte-verified against the SWH archive. Citation-grade qualified SWHIDs preserved.

MapIt.pm · 571 B · ext .pm · seen 12383x in SWH

[via heuristic](#) [rule h/linguist/pm/0](#)

swh:1:cnt:238c5c62c8afb3beb1373476b9857b7af32834d;origin=https://github.com/mysociety/fixmystreet;anchor=swh:1:rev:c525f578acef784e08624d1444feff7949b9ace;path=/perl/lib/FixMyStreet/MapIt.pm

[Open in SWH](#) · [Raw bytes \(SWH\)](#) · [GitHub raw](#)

[Show source](#)

98\_STOCKQUOTES.pm · 20515 B · ext .pm · seen 11349x in SWH

[via heuristic](#) [rule h/linguist/pm/0](#)

swh:1:cnt:425464fa7112720ba04fc8d3f525b9de08bde9c;origin=https://github.com/rhop/them-mirror;anchor=swh:1:rev:52ca3800ed14f5e930f7e7d9163ae3bc77aae8b;path=/them/contrib/98\_STOCKQUOTES.pm

[Open in SWH](#) · [Raw bytes \(SWH\)](#) · [GitHub raw](#)

[Show source](#)

resultset\_overload.t · 543 B · ext .t · seen 1858x in SWH

[via heuristic](#) [rule h/linguist/t/0](#)

swh:1:cnt:c5ecce87432c0b1e46520cb9b15983f08dc302;origin=https://github.com/st3fan/osx-10.9;anchor=swh:1:rev:9f8135f87c2ba27b1e76a514464b023812eff645;path=/CPAN/Internat-148/DBIx-Class/t/resultset\_overload.t

[Open in SWH](#) · [Raw bytes \(SWH\)](#) · [GitHub raw](#)

[Show source](#)

IssuingRules.pm · 7958 B · ext .pm · seen 1695x in SWH

[via heuristic](#) [rule h/linguist/pm/0](#)

swh:1:cnt:6023671522ebe2915fc8348a5353e5d2dab14084;origin=https://github.com/laurenthd/koha-bibliolibre;anchor=swh:1:rev:489b948a84a9efeba23e983ca86942cc2ec439d;path=/C4/IssuingRules.pm

Real programs from SWH: qualified SWHID + the rule that attributed them

#### Extension review queue

Ranked queue of file extensions present in Software Heritage that need a label. Sorted by priority — highest = most files in SWH and least already claimed by a PL in our taxonomy.

4,556  
in queue  
(popular ≥ 10.0K occurrences & not well-attributed)

2,805  
orphans  
(zero evidence anywhere)

1,223  
popular exts skipped  
(≥ 10.0K occurrences & already well-attributed)

1,803  
well-attributed exts  
(whole taxonomy, any popularity)

1,994  
total exts  
in taxonomy

11  
manual labels  
submitted so far

| Extension        | SWH   | Last | #Claim | Current claimants | Auto-suggested                 | Samples | Orphan? | Action                    |
|------------------|-------|------|--------|-------------------|--------------------------------|---------|---------|---------------------------|
| .map, .MAP       | 87.8M | 2023 | 0      | —                 | <a href="#">Build artifact</a> |         |         | <a href="#">help this</a> |
| .log, .Log, .LOG | 84.2M | 2023 | 0      | —                 | <a href="#">Build artifact</a> |         |         | <a href="#">help this</a> |
| .lock            | 61.4M | 2023 | 0      | —                 | <a href="#">lock/cache</a>     |         |         | <a href="#">help this</a> |
| .cache, .Cache   | 46.6M | 2023 | 0      | —                 | <a href="#">lock/cache</a>     |         |         | <a href="#">help this</a> |
| .info, .INFO     | 46.5M | 2023 | 0      | —                 | <a href="#">Build artifact</a> |         |         | <a href="#">help this</a> |
| .flat            | 39.3M | 2023 | 0      | —                 | <a href="#">unknown</a>        |         |         | <a href="#">help this</a> |

A review queue: 4,556 popular extensions waiting for a label

# What the encyclopedia offers

Features of the current prototype

## Browse & search

- ▶ 14,242 language pages, A–Z, search by name or alias
- ▶ filters: has an SWH program, has an LLM program, number of sources
- ▶ one page per source: what is only in Esolang?

## A page per language

- ▶ which sources attest it
- ▶ Wikipedia / Wikidata facts: designer, year, influences
- ▶ claimed extensions, with source, strength and SWH counts
- ▶ real programs from SWH: qualified SWHID + the rule that fired

## Tools & agents, three jobs

- ▶ **find languages**: 7 catalogs, Wikidata candidates, LLM agents
- ▶ **label programs**: PLI tools (Linguist rules, Pygments, Synid), LLM judges, human reviews
- ▶ **find programs**: SWH mining per extension, sample requests

## A page per extension

- ▶ claimants and attribution status (well / weakly attributed, polysemous, orphan)
- ▶ SWH popularity, case variants (`.R` vs `.r`)
- ▶ Linguist disambiguation rules

## Contribute & review

- ▶ add a language, add a program, label an extension, request SWH samples
- ▶ forms → GitHub issues → merged, auditable
- ▶ review queue, candidates from Wikidata (391)

## Transparency

- ▶ every fact is a file in git, every claim has a source
- ▶ statistics: sources, models, agents
- ▶ recent submissions, one-line revert

# The dream: every language, with its real programs

The big picture

A **comprehensive, living encyclopedia** of programming languages, **together with their programs in Software Heritage**, from ENIAC to last week's DSL.

- ▶ **Exhaustive**: history included (HOPL, Esolang, PLDB, Linguist...)
- ▶ **Anchored in code**: citable SWHIDs, or a reasoned evidence of absence
- ▶ **Traceable**: every claim says who asserts it, and why
- ▶ **Living**: humans, AIs and automatic procedures add, correct, validate
- ▶ **Sustainable**: carried by SWH × CodeCommons, not a side project

*Every language with its programs, every program with its language.*

## COBOL 1959 · CODASYL

🕒 **Lineage** FLOW-MATIC, COMTRAN, FACT... → COBOL → PL/I (HOPL)

✓ **Sources** PLDB · Linguist · Pygments · Wikidata

✓ **Extensions** .cob .cbl .CBL .cpy, case matters

✓ **Programs** ≈150,600 genuine files, 6,278 repos, SWHIDs

🕒 **Usage over time** raw counts, COBOL extensions



○ **Dialects, domains, papers** at archive scale: to build

✓ available today · 🕒 partial · ○ to build

# Why now, and why it matters

## ✓ The right time

- ▶ **Software Heritage exists:** the largest archive of source code, with citable identifiers, and CodeCommons to analyse it at scale. A unique opportunity
- ▶ **The technology is here:** PLI tools (Linguist, Synid, Tree-sitter), LLMs to propose and annotate, review and collaboration tools
- ▶ → we can map every language to its programs, and every program to its language

## ✗ The status quo

- ▶ **Many ad-hoc attempts:** HOPL, PLDB, Esolang, Wikipedia lists, Linguist, Rosetta Code... each with its own idea of "a language"
- ▶ **Loosely connected:** no shared identifiers, names that don't match, no link to real code
- ▶ **Fragile:** one-person efforts freeze or go offline (HOPL: frozen around 2005, often down)

### Heritage

what ALGOL, COBOL or Icon code really looked like

### Science of programming

adoption, evolution and death, per language

### AI & LLMs

benchmarks and data for low-resource languages

### Tools & SWH

search and statistics by language; ground truth for PLI

### Legacy & industry

where is the COBOL, RPG, PL/I code, in which dialects?

### Education & citation

real, citable programs (SWHIDs) in papers and textbooks

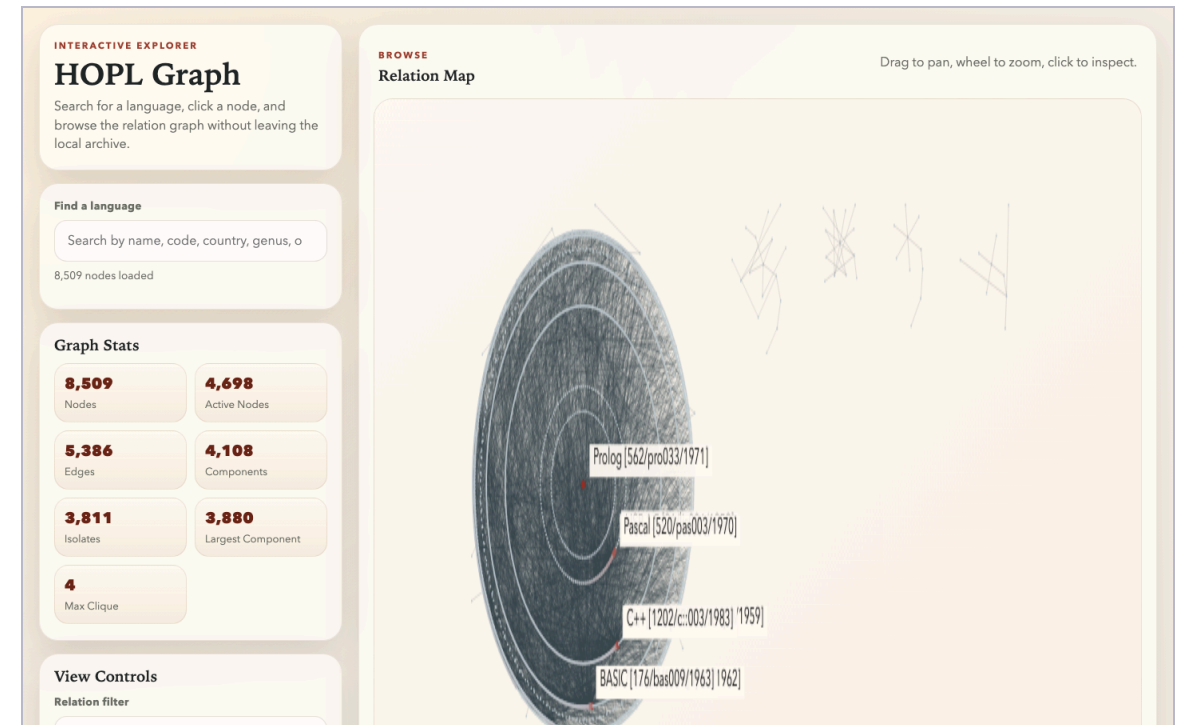
# 1. Four mirrors, no code

HOPL · PL-ultimate · PL-ultimate-llm · SWH file extensions

# HOPL: a genealogy, frozen in 2005

Mirror 1 · history

- ▶ [hopl.info](http://hopl.info): *the* History of Programming Languages, but often **down**, and "please do not copy"
- ▶ Scraped **politely** with Codex: one request at a time, long random delays, contact e-mail in the user agent, resumable, raw HTML archived
- ▶ → **8,509 languages**, **5,386 edges**, 165 relation labels (*Evolution of, Influence, Dialect of, Compiled to...*)
- ▶ Most connected: **Prolog**, FORTRAN IV, PL/I, LISP 1.5
- ▶ 1 language dated after 2004: no Rust, Go, Swift, Kotlin
- ▶ No code



| Source        | #     | Its view of "a language"      |
|---------------|-------|-------------------------------|
| Esolang       | 6,797 | esoteric, experimental, jokes |
| PLDB          | 5,288 | curated metadata              |
| Linguist      | 878   | detectable on GitHub          |
| Wikipedia     | 769   | encyclopedic article          |
| Hyperpolyglot | 670   | Linguist, in Rust             |
| Pygments      | 645   | someone wrote a lexer         |
| Rosetta Code  | 640   | solved toy tasks              |

# 12,608

entities in the union (11,963 in the first build;  
+ source updates, Wikidata)

- ▶ **88%** appear in **one** source only  
(Esolang 6,366, PLDB 4,034)
- ▶ Near-duplicates survive: `Python` / `py` / `Python` (programming language)
- ▶ Extensions for only ~**10%** of them

# PL-ultimate-llm: LLMs contribute through commits

Mirror 3 · what LLMs know

**One turn = one commit:** read the list, propose a real language **not yet in the list**, give an evidence URL and a program; git hooks validate; trailers record which model did it.

3,862 languages

3,846 commits

since Oct. 2025

Claude 3,392

Grok 246

Gemini 144

Llama 48

GPT 32

DeepSeek 9

|                              | in the LLM list | not in it |
|------------------------------|-----------------|-----------|
| in $\geq 1$ of the 7 sources | 2,439           | 10,400    |
| in none of them              | 1,403           | —         |

= the 14,242 pages. Icon sits in the top-left cell: PLDB, Pygments and Rosetta Code had it, the LLM list did not.

**Do LLMs invent languages? Rarely:** checking all 3,842 evidence links leaves  $\approx 23$  invented names (0.6%); broken evidence is far more common (appendix).

Commit 1a1c557

FAMILIAR-project committed on Oct 22, 2025

turn: add Icon (+1 program)

List-Digest: 95f48d04

Model: meta-llama/llama-3.1-70b-instruct

Temperature: 0.4

main

7 files changed

data/pl\_list.txt

@@ -53,6 +53,7 @@ HCL

53 53 Hoon

54 54 Humbug

55 55 Hy

56 + Icon

56 57 Idris

57 58 Io

58 59 J

languages/Icon/meta.json

@@ -0,0 +1,8 @@

1 + {

2 + "name": "Icon",

3 + "aliases": [

4 + "Icon Programming Language"

5 + ],

6 + "evidence\_url": "https://en.wikipedia.org/wiki/Icon\_(programming\_language)",

7 + "added\_at": "2025-10-22T15:15:16Z"

[github.com/acherm/PL-ultimate-llm/commit/1a1c557](https://github.com/acherm/PL-ultimate-llm/commit/1a1c557)

Roberto's CSV (SWH-MSR-ARV: Desmazières, Di Cosmo, Lorentz, MSR 2025): **2.96 M** extensions, occurrences per year

| Extension                              | Occurrences in SWH | Claimed by                                                        |
|----------------------------------------|--------------------|-------------------------------------------------------------------|
| <code>.m</code>                        | 46.5 M             | 12 claimants: Octave/MATLAB, Objective-C, Mercury, m4, Wolfram... |
| <code>.R</code> · <code>.r</code>      | 21.5 M · 1.7 M     | R (case-folding once lost 21.5 M files)                           |
| <code>.CBL</code> · <code>.cbl</code>  | 192 K · 57 K       | COBOL                                                             |
| <code>.icn</code> · <code>.icon</code> | 32 K · 19 K        | Unicon · Icon (Pygments)                                          |
| <code>.fsf</code>                      | 18 K               | claimed by no language (in fact an embedded DSL)                  |

**Extensions, not languages. A count is not a program.**

# Four mirrors, one blind spot

|                 | names languages | lineage       | extensions     | real programs        | maintained      |
|-----------------|-----------------|---------------|----------------|----------------------|-----------------|
| HOPL            | ✓ 8,509         | ✓ 5,386 links | ✗              | ✗                    | ✗ frozen ~2005  |
| PL-ultimate     | ✓ 12,608        | 🟡 infoboxes   | 🟡 10% of names | ✗ toy tasks          | ✓ upstream      |
| PL-ultimate-llm | ✓ 3,862         | ✗             | ✗ rarely       | 🟡 generated          | ✓ agents, forms |
| SWH extensions  | ✗ suffixes only | ✗             | ✓ 2.96 M       | 🟡 real, unattributed | ✓               |

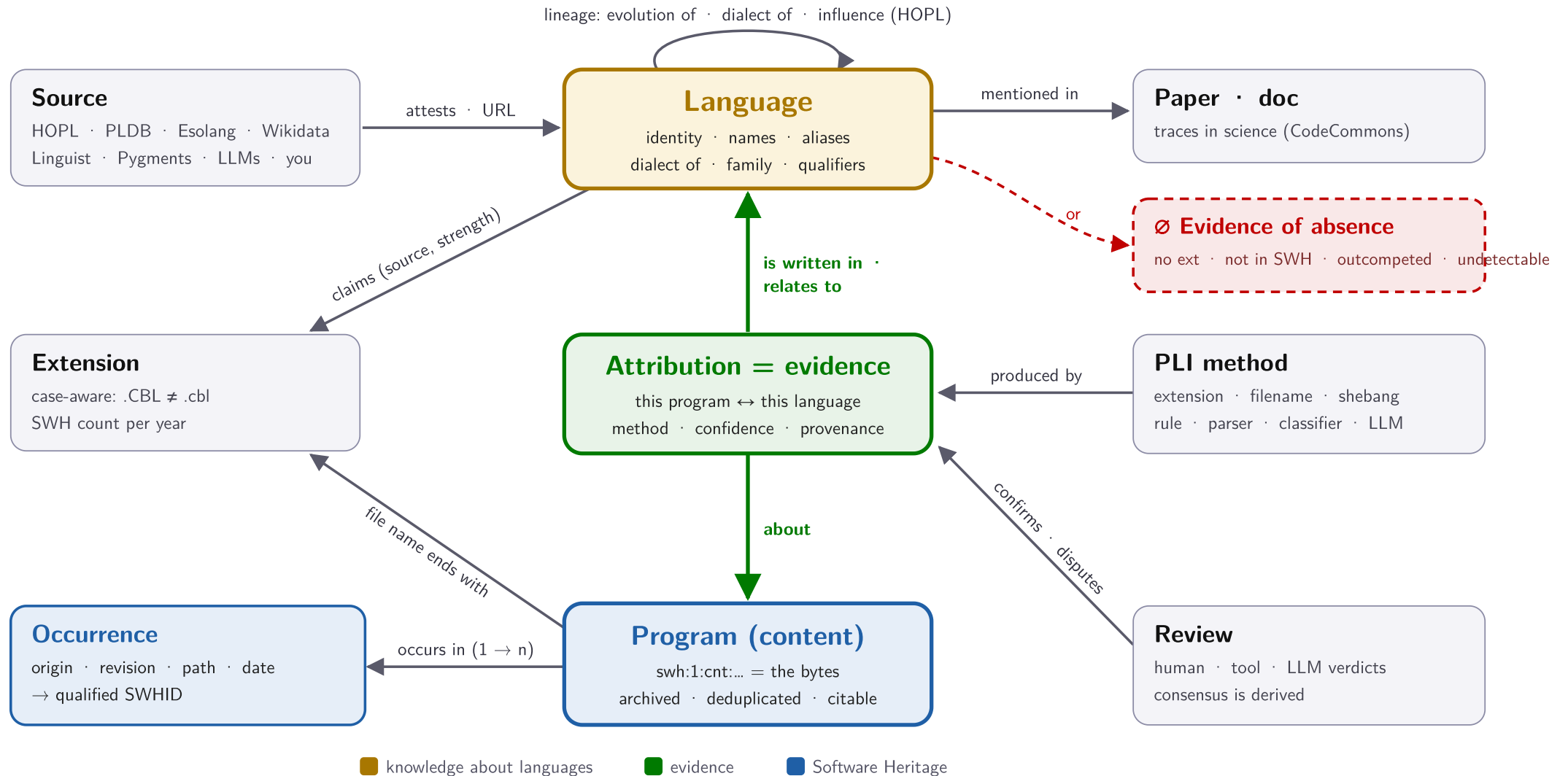
No row has both **names languages** and **real programs**: nothing links a language to the programs that exist.  
"Give me 10 real, non-toy **Mercury** programs, with citable provenance": today, impossible.

## 2. The vision, made concrete

An ontology, a prototype, and the identification problem

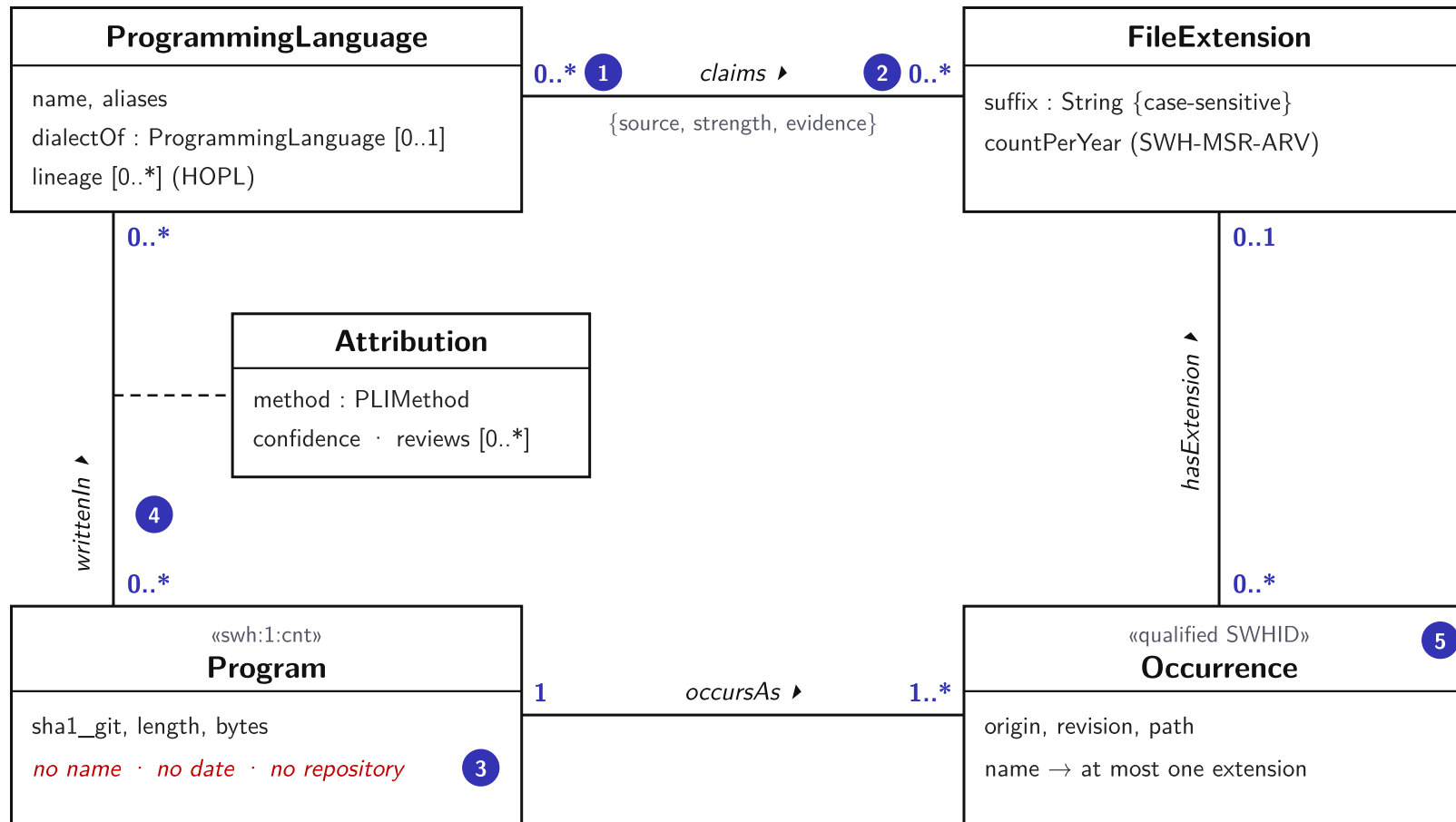
# Languages, evidence, programs

## The ontology



# PL, extension, program: it's all in the cardinalities

A UML view of the core concepts, as seen from Software Heritage



**1 A language claims 0..\* extensions:**  
~90% of catalogued languages claim none (esoteric, historical, no files); a few claim 10+ (Python, Ruby).

**2 An extension is claimed by 0..\* languages:** 0 for most (88% of extensions with  $\geq 1,000$  files in SWH: `.map`, `.log...`); \* is polysemy (`.m`: 12 claimants).

**3 Names live in directories:** an SWH program is just bytes, so the extension belongs to each occurrence, and one program occurs in many places.

**4 writtenIn is established per program,** with a method and a confidence, never inferred from a claim: 46% of `.cbl`/`.CBL` files are not COBOL.

**5 Counting = choosing a class:** programs (by file) or origins (by repo); a new revision is a new program.

# Two gaps, in numbers

Languages without extensions, extensions without languages

## Languages without extensions

**90%** of the 12,608 catalogued languages (11,366) claim **no** file extension: invisible to extension-based detection

## Extensions without languages

**88%** of the 18,120 extensions with  $\geq 1,000$  files in SWH (15,957) are claimed by **no** language; 56% of those with  $\geq 100,000$  files

Help label them: `/review/extensions/` lists **4,556** popular extensions ( $\geq 10,000$  files) without a claimant, 2,805 of them with no evidence at all. **11** labels so far.

### Extension review queue

Ranked queue of file extensions present in Software Heritage that need a label. Sorted by priority — highest = most files in SWH and least already claimed by a PL in our taxonomy.

4,556

in queue

(popular ≥ 10.0K occurrences & not well-attributed)

2,805

 orphans

(zero evidence anywhere)

1,223

popular exts skipped

(≥ 10.0K occurrences & already well-attributed)

1,803

well-attributed exts

(whole taxonomy, any popularity)

1,994

total exts

in taxonomy

11

manual labels

submitted so far

| Extension                     | SWH   | Last | #Claim | Current claimants | Auto-suggested            | Samples                                                                                        | Orphan?                                                                                                 |
|-------------------------------|-------|------|--------|-------------------|---------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| <code>.map, .MAP</code>       | 87.8M | 2023 | 0      | —                 | <div>build-artifact</div> |                                                                                                |                                                                                                         |
| <code>.log, .Log, .LOG</code> | 84.2M | 2023 | 0      | —                 | <div>build-artifact</div> |                                                                                                |                                                                                                         |
| <code>.lock</code>            | 61.4M | 2023 | 0      | —                 | <div>lock/cache</div>     |                                                                                                | <div> orphan</div>   |
| <code>.cache, .Cache</code>   | 46.6M | 2023 | 0      | —                 | <div>lock/cache</div>     | <div></div> |                                                                                                         |
| <code>.info, .INFO</code>     | 46.5M | 2023 | 0      | —                 | <div>build-artifact</div> |                                                                                                |                                                                                                         |
| <code>.flat</code>            | 39.3M | 2023 | 0      | —                 | <div>unknown</div>        |                                                                                                | <div> orphan</div> |

# PLI, formally: learn $f$ , one iteration at a time

## Conceptualization

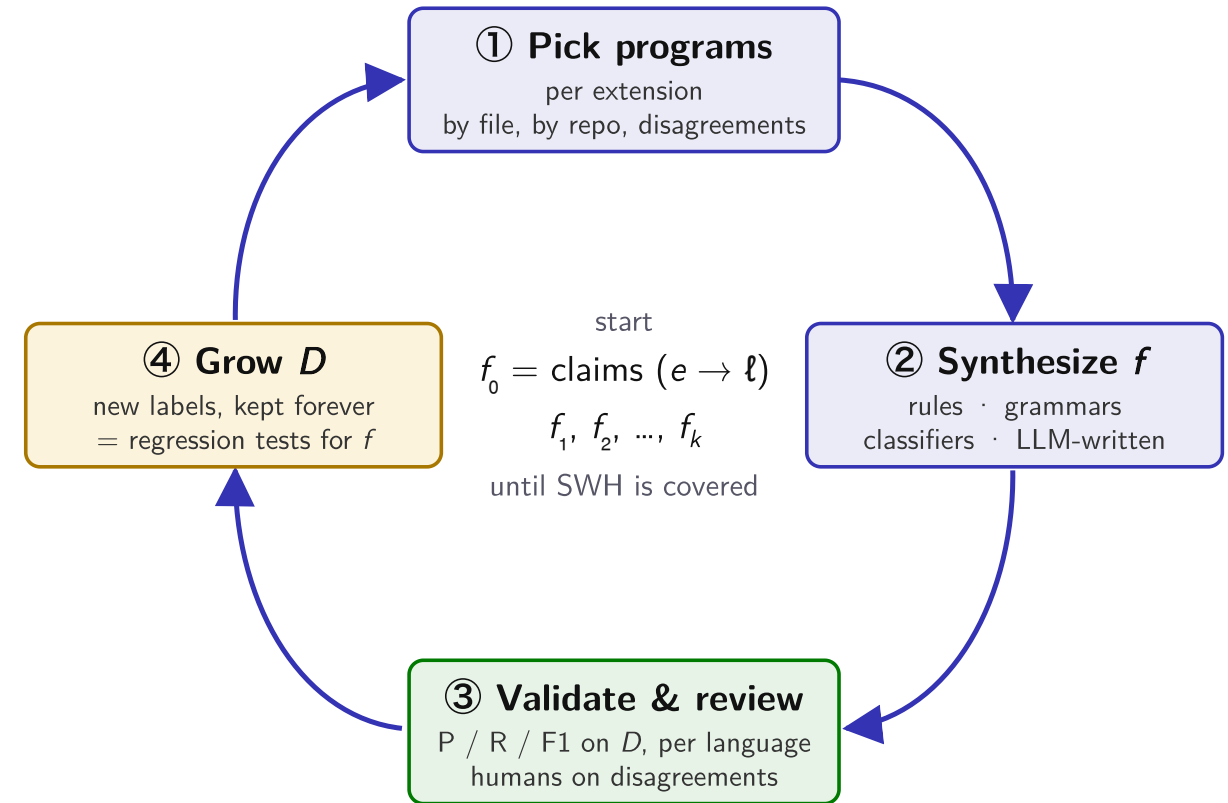
### Definition (programming language identification)

- ▶ **Languages**, the labels:  $L = \{\ell_1, \dots, \ell_n\}$ ,  $n \approx 12,600$ , plus  $\perp$  (not a language, unknown)
- ▶ **Extensions**:  $E = \{e_1, \dots, e_m\}$ ,  $m \approx 2.96$  M in SWH
- ▶ **Programs**:  $P$ , the SWH contents;  $p$  occurs with extensions  $\text{ext}(p) \subseteq E$
- ▶ **Claims**:  $C \subseteq L \times E$ , prior knowledge (catalogs, PLI tools)

**Goal**: find  $f: P \times E \rightarrow 2^L$  such that  $f(p, e) = \text{writtenIn}(p)$ : usually one  $\ell$ ,  $\emptyset$  for  $\perp$ , several for embedded languages

**Knowledge**: labelled examples  $D = \{(p, e, \ell)\}$ , claims, rules, grammars, models

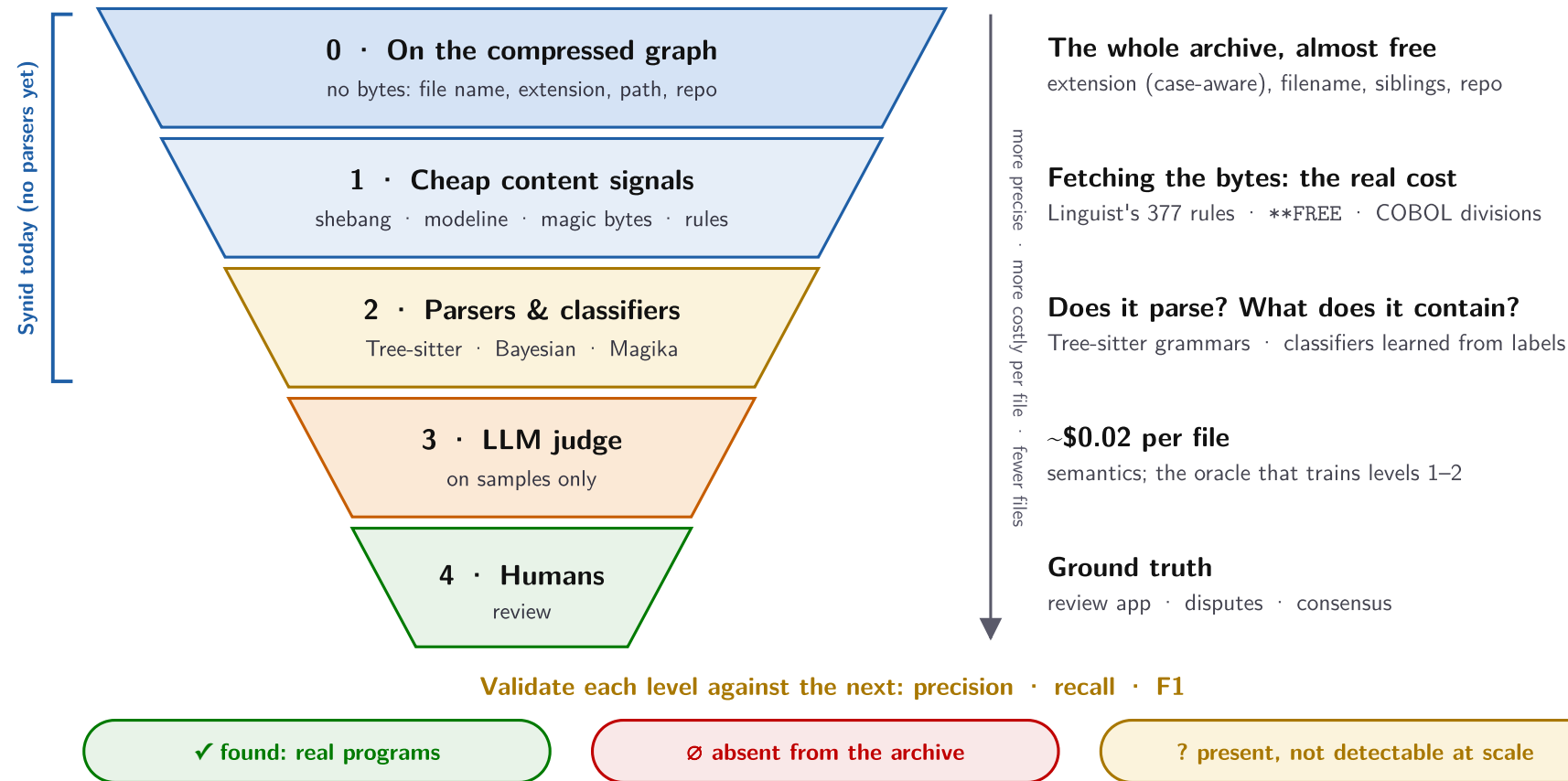
**Quality**: precision, recall, F1 of  $f$  on  $D$ , per language and per extension



Not a chicken-and-egg problem (programs to learn  $f$ ,  $f$  to find programs), but an **iterative process**: bootstrap from claims and a few labelled programs; every validated label becomes a **regression test**; examples and knowledge accumulate **across all of SWH**.

# Which language is this file in? A multi-level funnel

## Programming language identification (PLI)



PLI tools know ~**1,100** syntaxes, Tree-sitter several hundred grammars; we catalogue ~**12,600** languages, ~**90%** with no known extension. Difficult, and fascinating.

**Ongoing work**, DiverSE team (Rennes): Caroline Landry, Mathieu Acher, Axel Amour N'cho, Baptiste Mehat, Bignon Lokonon, Corentin Ollivier, Guillaume Claudic, Stephan Kunne

### 3. Zoom in

What is *actually* in file extension X, in all of Software Heritage?

# One extension, all of it

A complementary, MSR-style contribution

**Why?** Low-resource languages:

- ▶ understand **real usage** and the ecosystem
- ▶ build **benchmarks, fine-tune** LLMs
- ▶ find **interesting repositories**

**Input** (thanks Roberto, Valentin, Baptiste): *every* SWH file with extension X + **its origin**, extracted from the graph with CodeCommons tools

SWH » GitHub, but reading it takes method.

1. **Sample**: by file *and* by repository
2. **Fetch** bytes by SWHID (cached)
3. **Indicators**: cheap, deterministic
4. **LLM-as-judge**: enum-constrained
5. **Reclassifier**: zero-cost, validated vs. the judge
6. **Origins**: provenance, concentration
7. **Human review**: all labels side by side

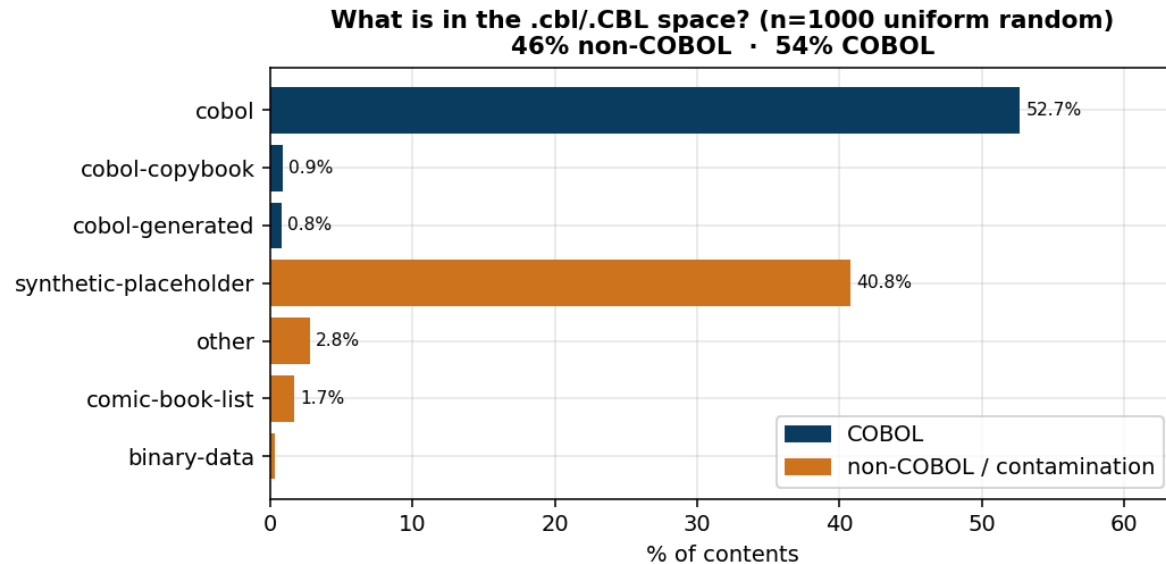
~70% of the toolkit is reused from one extension to the next

# Three extensions, three stories

|                       | <code>.cbl</code> · <code>.CBL</code>           | <code>.fsf</code>                             | <code>.rpgle</code>                      |
|-----------------------|-------------------------------------------------|-----------------------------------------------|------------------------------------------|
| unique files · repos  | 276,831 · 6,278                                 | 21,802 · 757                                  | 13,002 · 534                             |
| really the language?  | 54% COBOL                                       | an <b>embedded DSL</b> (FSL FEAT, in Tcl)     | 99.5% ILE RPG                            |
| what else is in there | one GitLab test fixture (40%), comic-book lists | GLSL shaders, DataLad pointers                | parsers <i>for</i> RPG: 21% of all files |
| the story             | contamination, <b>case matters</b>              | <b>fMRI</b> analysis designs; <b>polysemy</b> | the <b>sampling frame</b> decides        |

Next: `.m` (MATLAB/Octave/...), and any extension you care about.

# Lesson 1: an extension is a claim, not a fact



- ▶ **46%** of `.cbl`/`.CBL` files are **not COBOL** (95% CI 42.5–48.7)
- ▶ ~110,000 files say *"This is cobol file number N"*: one GitLab repo, a test fixture for huge trees
- ▶ `.CBL`  $\neq$  `.cbl`: mainframe enterprise vs learners (+ Calibre comic-book lists)
- ▶ ...yet **93%** of the repositories do contain COBOL

**/WBC\_97464\_FOO.CBL**

sw:1:cmt:4c699ec09243ebef9b1577805c11e422f24977 · 32 bytes · 1k · SWH ↗

```
This is cobol file number 97464
```

**ORIGIN**

graph  
<https://gitlab.com/tbetestpublic/repo-with-many-files-in-one-tree.git> gitlab.com  
source/WBC\_97464\_FOO.CBL · changing-commits · 2022-01-15T13:51:44Z SWH ↗

**LABELS (2 IS-COBOL VOTES · UNANIMOUS)**

|               |     |                       |
|---------------|-----|-----------------------|
| reclassifier  | non | synthetic-placeholder |
| division-gate | non | n_divisions=0         |

## 4. Making it real

An official, sustainable SWH × CodeCommons initiative

# Contributions: trace everything, keep it discussable

Today's workflow is a prototype; the principle is what matters

| Contribution          | How                                                                     | Lands as                                          |
|-----------------------|-------------------------------------------------------------------------|---------------------------------------------------|
| Propose a language    | web form → GitHub issue <code>pl-add</code>                             | PR, <b>merged on arrival</b>                      |
| Label an extension    | form on <code>/ext/&lt;x&gt;/</code> → issue <code>ext-review</code>    | accepted by default → <code>ext_claim</code>      |
| Ask for real programs | "Request SWH samples" → issue <code>sample-request</code>               | targeted SWH mining                               |
| Submit one file       | <code>submit_sample.py</code> , SWHID checked with <code>/known/</code> | <code>samples/&lt;pl&gt;/&lt;sha1_git&gt;/</code> |
| Review a program      | review UI, blinded; reviewer = human · tool · LLM                       | one immutable JSON per review                     |
| AI agents             | parallel campaigns in git worktrees                                     | one commit per language                           |

Today: forms → GitHub issues → pull requests, accepted by default, reverted if wrong. **The UI and the workflow are clunky, and will be redesigned.**

## What to keep

- ▶ **every contribution is a traced fact:** who, when, why, with which evidence
- ▶ **open to discussion:** each claim has a thread; disagreement is kept, not overwritten
- ▶ **revertible;** humans, tools and LLMs on the same footing
- ▶ facts as files, aggregates derived

# Challenges: what's missing

## Programming language identification (PLI) at archive scale: the central challenge

- ▶ **Coverage:** PLI tools know ~1,100 syntaxes (Linguist, Pygments, Synid), Tree-sitter several hundred grammars; we catalogue ~12,600 languages, 90% without an extension
- ▶ **Ambiguity:** 44 extensions have 6+ claimants (`.m`, `.h`, `.pl`, `.t`); ~70 with several primary claimants have no disambiguation rule
- ▶ **No ground truth:** no benchmark measures PLI tools on SWH, per language; reviews and extension studies can build one
- ▶ **Scale:** billions of files: cheap filters on the compressed graph, costly ones on content, every level validated (P / R / F1)

## Identity

a canonical language ID authority, an "ORCID for languages"; merge `Python` / `py` / ...; bring HOPL's genealogy in

## SWH-native evidence

per-extension **extractors** (every file + origin, from the SWH graph) are costly: CodeCommons tools, so far run for `.cbl`, `.fsf`, `.rpgle`; then a **complete mapping** of SWH programs (a laptop scan: killed after 6.8 h)

## Evidence cards

for all ~12,600 languages, with a formal evidence-of-absence verdict

## Governance & hosting

who curates? what counts as a language? data license? Today: one person's repo + GitHub Pages

# How you can help, concretely

## SWH / CodeCommons infrastructure

- ▶ run the CodeCommons extractors on more extensions (as for `.cbl`, `.fsf`, `.rpgle`), towards a complete mapping
- ▶ a per-extension index in the derived datasets
- ▶ graph / provenance access; a neutral home

## PL / ML / MSR researchers

- ▶ PL identification at archive scale: Tree-sitter grammars, classifiers, a PLI benchmark built from SWH
- ▶ sampling & labeling methodology
- ▶ benchmarks, fine-tuning data for low-resource languages

## Domain experts: adopt an extension

- ▶ COBOL, neuro `.fsf`, `.m`, IBM RPG... or yours
- ▶ dive deep: define the taxonomy, review, tell the story

## Everyone, one click

- ▶ add a language · label an extension
- ▶ review a program · flag a hallucination

# Catch them all!

Gamifying the encyclopedia: programs and languages

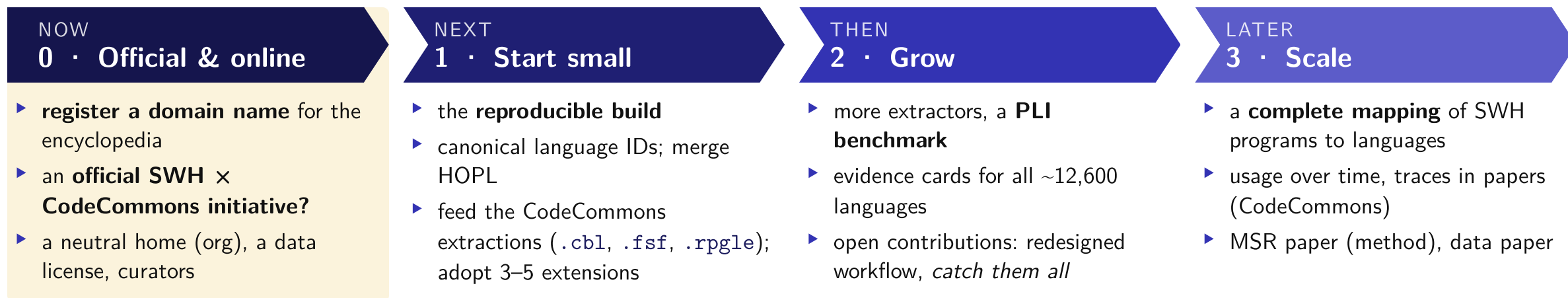
|                                                                                        |                                                                                 |                                                                            |                                                                                 |
|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <div>●</div> <div><b>COBOL</b></div> <div>2 on the site ·<br/>≈150K in the study</div> | <div>●</div> <div><b>Perl</b></div> <div>caught · 6<br/>programs</div>          | <div>●</div> <div><b>Pascal</b></div> <div>caught · 5<br/>programs</div>   | <div>●</div> <div><b>Mercury</b></div> <div>caught · 1<br/>program</div>        |
| <div>○</div> <div><b>C</b></div> <div>seen · not yet<br/>caught</div>                  | <div>○</div> <div><b>Fortran</b></div> <div>29 variants · none<br/>caught</div> | <div>○</div> <div><b>Icon</b></div> <div>5 catalogs · 0<br/>programs</div> | <div>○</div> <div><b>Smalltalk</b></div> <div>6 sources · 0<br/>programs</div>  |
| <div>○</div> <div><b>Brainfuck</b></div> <div>6 sources · 0<br/>programs</div>         | <div>○</div> <div><b>ALGOL 60</b></div> <div>0 programs</div>                   | <div>○</div> <div><b>Porth</b></div> <div>found by an LLM</div>            | <div>✗</div> <div><del><b>BASIC-D</b></del></div> <div>invented by an LLM</div> |

192 of 12,608 languages caught (1.5%). A **pipeline gap** more than an archive gap: extracting every file of an extension from SWH is costly; CodeCommons extractors did it for `.cbl`, `.fsf`, `.rpgle`, not yet for the others.

- ▶ **Catch** a language: attach its first verified program from SWH; every catch is reviewed, invented languages get caught out
- ▶ **Power-ups**: each per-extension extractor (CodeCommons) catches a whole extension at once: `.cbl/.CBL` alone yields ≈150,600 COBOL files
- ▶ **Rarity**: the fewer programs in the archive, the rarer the catch
- ▶ **Quests**: catch C and Fortran (still missing!), label `.flat`, check the 465 unverified LLM names
- ▶ **Collections**: a family (ALGOL's descendants), a decade, an extension
- ▶ **Leaderboards**: for people, and for models (the LLM game already counts 18)

# Roadmap: an official SWH × CodeCommons initiative?

Start with a name and a home, then grow incrementally



**Throughout: one reproducible pipeline.** Inputs of any kind (catalogs, HOPL, SWH datasets, PLI tools, LLMs, **human insights and reviews**) → site, datasets, evidence cards; rebuilt from scratch at any time, every number traceable to its inputs.

## Decisions for today

**Who's in? · Which domain name, which home? · Which extensions do we adopt first?**

# Towards an encyclopedia of programming languages anchored in Software Heritage & CodeCommons

*Every language with its programs, every program with its language.*

From the beginning of computing to today, with Software Heritage as the source of truth for the code.

PL Catalog: [blog.mathieuacher.com/PL-ultimate-llm](http://blog.mathieuacher.com/PL-ultimate-llm) · [github.com/acherm/PL-ultimate-llm](https://github.com/acherm/PL-ultimate-llm)

Reports: [docs/cobol\\_swh\\_study.pdf](#) · [fsf\\_swh\\_study.pdf](#) · [rpgle\\_swh\\_study.pdf](#)

Thanks to Software Heritage (Thomas Aynaud, Roberto Di Cosmo, Stefano Zacchiroli, Valentin Lorentz) for feedback, assistance and the SWH extractions; to the DiverSE team (Rennes) and the PLI group (Caroline Landry, Axel Amour N'cho, Baptiste Mehat, Bignon Lokonon, Corentin Ollivier, Guillaume Claudic, Stephan Kunne); and to everyone who will adopt an extension.



# Appendix

**Icon** (Griswold, 1977): generators, goal-directed evaluation, string scanning.

- ▶ **HOPL** knows its family: 21 relations (SNOBOL4, SL5, Unicon, Python...)
- ▶ **5 catalogs** list it (PLDB, Pygments, Wikipedia, Rosetta Code, Wikidata)
- ▶ **An LLM** (Llama 3.1 70B) added it, with a program that defines `main` three times
- ▶ **0** verified Icon program in the catalog so far

|                        | <code>.icn</code>                                                          | <code>.icon</code>                                                                                                     |
|------------------------|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| files in SWH           | 32,257                                                                     | 19,184                                                                                                                 |
| Pygments says          | Unicon                                                                     | Icon                                                                                                                   |
| what they really hold* | Icon <b>and</b> Unicon sources (the official Icon repository alone: ~1.2K) | mostly <b>not</b> Icon: icon-theme metadata, icon images; the Icon code comes mainly from <b>one</b> Rosetta Code dump |

\* GitHub code search, Sept. 2026: ~9.3K `.icn` files; ~56K `.icon` files, of which ~16.5K icon-theme metadata and ~1.3K with Icon code (667 in one Rosetta Code dump).

Finding Icon's programs needs **automated tools and supervised methods** (which extension? which content? Icon or Unicon?) that **scale to all of Software Heritage**: a challenge, and an opportunity.

# Appendix · Do LLMs invent languages? We checked the evidence

All 3,842 evidence links of the LLM list, tested in September 2026

- ▶ **2,439** (63%) are also in one of the 7 sources; of the **1,403** others, **277** are in HOPL or Synid
- ▶ **Evidence links are fragile**: **28%** of the Wikipedia links point to a page that does not exist (**40%** for LLM-only names); others land on the wrong page (Brython → *Ancient Celtic people*)
- ▶ A missing page ≠ a fake language: *Aardappel* (*programming language*) has no page, yet Aardappel exists
- ▶ **465** names stay unverified: broken or off-topic evidence, no corroboration

## Web check of 40 random unverified names

**36** are real languages: 27 as named, 9 under another name (Gringo, SPU assembly, ObScript...). Only their evidence link was wrong.

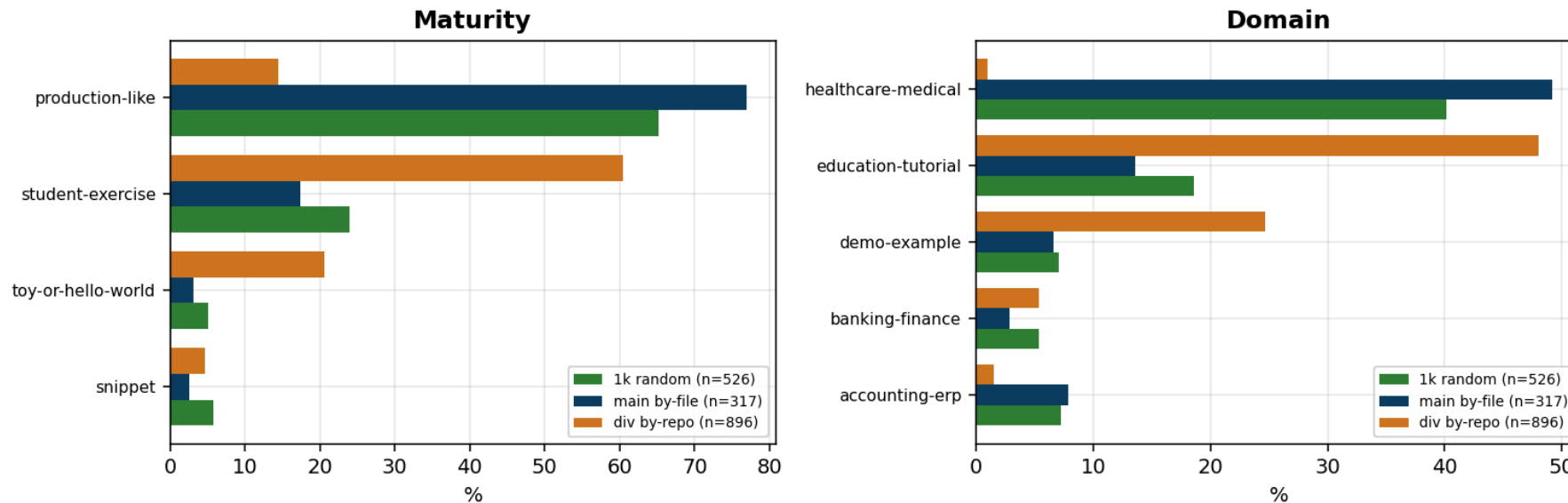
**2** are real, but not languages: NATJ (a Java binding library), Brython (a Python 3 implementation)

**2** not found, likely invented: BASIC-D, Oxymoron

≈ 5% of the 465 (95% CI 1–17%): about **23 invented names in 3,842** (0.6%, at most 2%).  
**LLMs invent evidence far more often than languages.**

# Appendix · Lesson 2: how you count changes the story

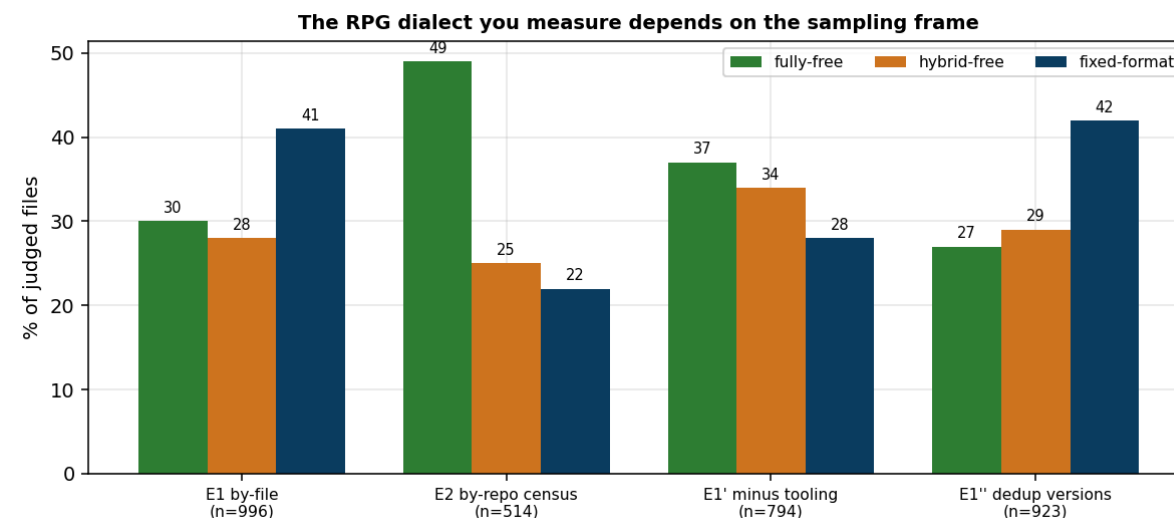
Sampling frame: random (file) vs curated by-file vs by-repo



- ▶ **by file:** 77% production-like, 49% healthcare (one system, JMA ORCA, hundreds of files)
- ▶ **by repository:** 14% production-like, **60% student exercises**, median 48 LOC (vs 596)
- ▶ **71 repositories** (1.1%) hold 80% of all COBOL-extension files
- ▶ "Typical **file**" ≠ "typical **project**": report both (same story for RPG, next)

## Appendix · .rpgle: a clean extension, a biased archive

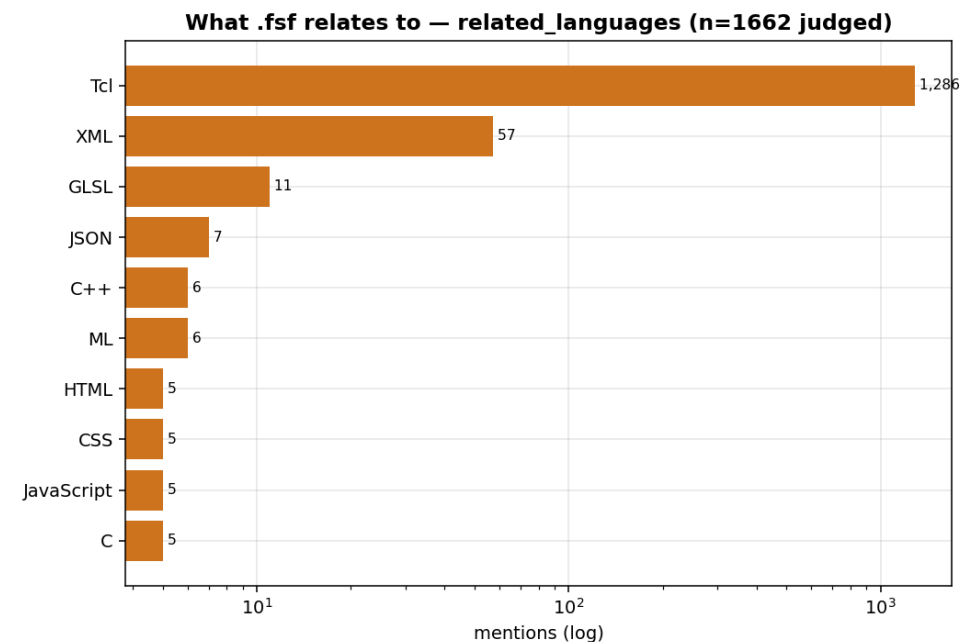
- ▶ **IBM RPG** (1959), still in production on IBM i;  
.rpgle = ILE RPG / RPG IV
- ▶ **13,002 files** · **534 repos** · **99.5%** genuine  
RPG: an honest extension
- ▶ One language, **three formats**: fixed (column-based), hybrid, fully-free (\*\*FREE on line 1)
- ▶ **Which RPG?** Fixed-format: **41%** by file, **22%** by repo. Fully-free: 30% → **49%**
- ▶ Why: three **parsers for RPG** (an interpreter, two ANTLR grammars) hold **21%** of all .rpgle, 92% fixed-format test fixtures
- ▶ File versions (35% of contents) do **not** bias: 41% → 42%



The archive's picture of RPG is shaped by the tools built to parse RPG.

# Appendix · Lesson 3: an LLM judge is a useful annotator, not an oracle

- ▶ **Enum-constrained** structured outputs (free text fragmented into 30+ spellings)
- ▶ Ask "what **is** this content, what does it **relate to?**", not "is it COBOL?"
- ▶ ~\$0.02 per file: a study costs tens of dollars. **All of SWH: no way**
- ▶ → bootstrap **cheap classifiers**: COBOL reclassifier P 1.00 · R 0.79 · F1 0.88 vs. the judge
- ▶ RPG: the judge contradicts the literal **\*\*FREE** directive on **9.1%** of files, one-sided (132 vs 2)
- ▶ **Code decides lexical facts, the LLM semantic ones, humans arbitrate**



# Appendix · From case studies to a method, and back

## An MSR contribution

**A general methodology** for "what is in extension X?"

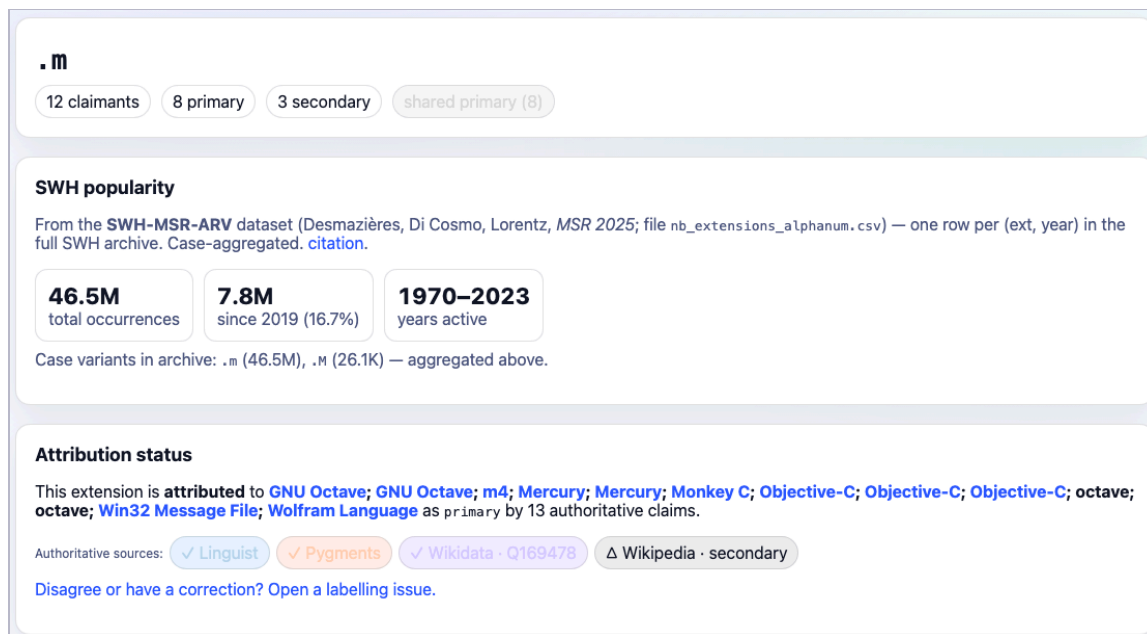
- ▶ **sampling**: by file, by repo, stratified... which question does each answer?
  - ▶ **annotating**: LLM vs. PLI tools vs. ad-hoc/ML classifiers; the label problem
  - ▶ **validating**, iterating until *all* files of an extension are covered
- + **domain** contributions: COBOL (batch, CICS...), neuroimaging, IBM i, `.m`

## What it gives the encyclopedia

- ▶ a **case-aware** extension → language mapping (`.CBL` ≠ `.cbl`, `.R` ≠ `.r`)
- ▶ **polysemy** made explicit: `related_languages`, `expressed_in`
- ▶ **validated cheap classifiers** → evidence at archive scale
- ▶ evidence of absence, **measured**

Encyclopedia = **breadth** (12K languages, a few programs each). Zoom-in = **depth** (one extension, its whole population).

# Appendix · polysemy: an extension is not a language



- ▶ **.m**: **46.5 M** files, **12 claimants**
- ▶ Linguist's **377** disambiguation rules, made **runnable**: "Mercury, because `:- module` matched"
- ▶ Then `XSetModifierMapping.m` (Xorg) is... **none of them**: an xts5 test-case format
- ▶ `.icn` → Unicon, `.icon` → Icon, but Icon sources are `.icn`

**Classify content. Keep the provenance of every claim.**

For each language, Software Heritage is the **source of truth for the code**:

## ✓ Evidence of existence

- ▶  $\geq K$  **real archived programs**
- ▶ each with a **qualified SWHID**: citable, permanent
- ▶ and **why** it was attributed: extension? heuristic rule? classifier? human?

## ∅ Evidence of absence

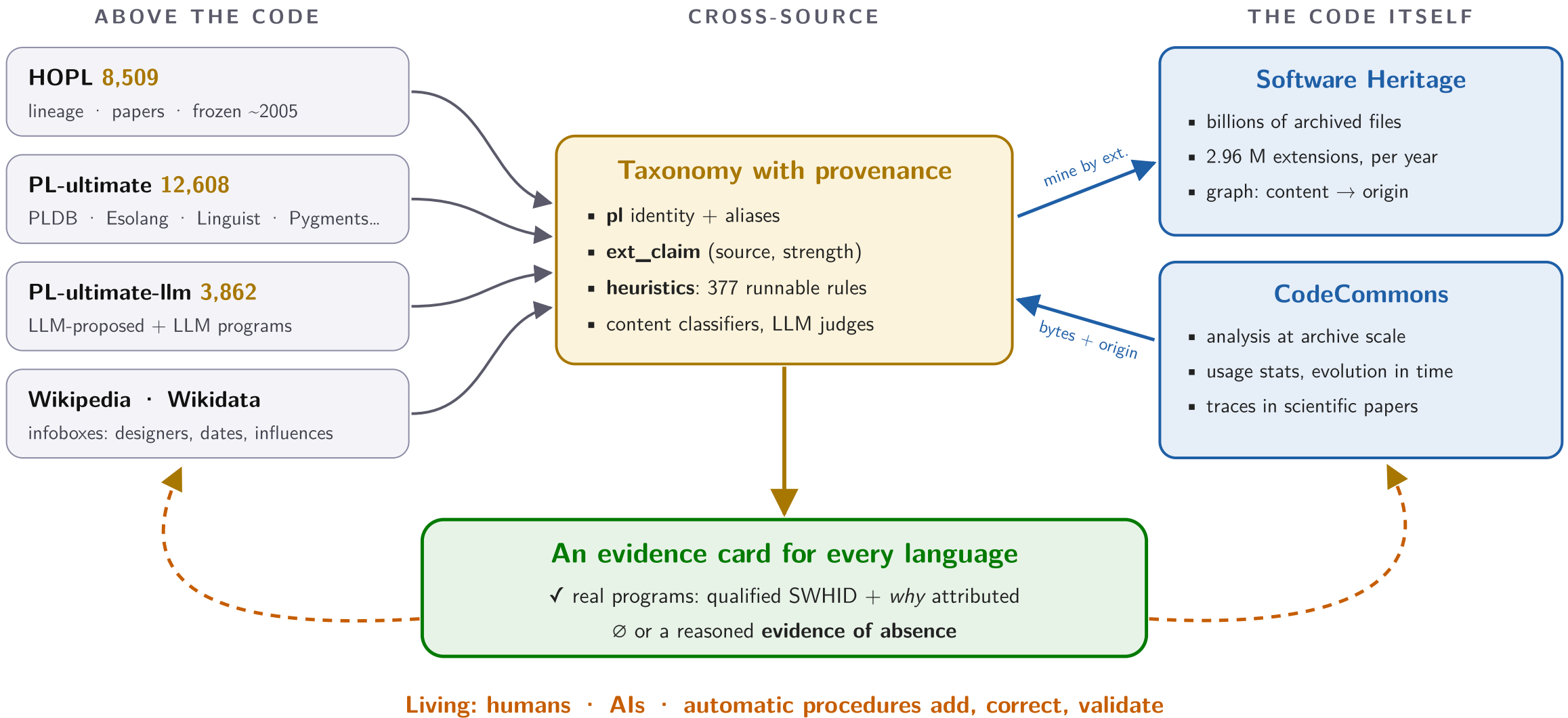
- ▶ no extension claimed anywhere
  - ▶ extension claimed, absent from SWH
  - ▶ files exist, but always won by a competitor (Mercury vs MATLAB on `.m`)
- A result in itself.

## ↻ Then, at CodeCommons scale

- ▶ usage statistics
- ▶ evolution over time: birth, peak, decline
- ▶ traces in scientific papers

**Living**: humans, AIs and automatic procedures add, correct, validate. Goal: **every language, from the beginning of computing to today**.

# Appendix · how the pieces fit



## Appendix · the landscape, source by source

| Source          | Size        | Captures                    | Limitation                        |
|-----------------|-------------|-----------------------------|-----------------------------------|
| HOPL            | ~8,500      | lineage, papers             | frozen ~2005, often down, no code |
| Esolang         | ~6,800      | esoteric, experimental      | many with ~zero usage             |
| PLDB            | ~5,100      | paradigms, dates, designers | sparse on real adoption           |
| Linguist        | ~800        | detection + ambiguity rules | GitHub's worldview                |
| Pygments        | ~600        | syntax highlighting         | exists if someone wrote a lexer   |
| Rosetta Code    | ~560        | tasks × languages           | toy snippets                      |
| PL-ultimate     | 12,608      | union of 7 (+ Wikidata)     | 88% single-source                 |
| PL-ultimate-llm | ~3,900      | LLM-proposed + programs     | hallucination risk                |
| SWH-MSR-ARV     | 2.96 M exts | occurrences per year        | extensions, not languages         |

```
swh:1:cnt:c5ecce87432c6b1e8465269cb9b15902fe0dc302
;origin=https://github.com/st3fan/osx-10.9
;anchor=swh:1:rev:9f8135f87c2ba27b1e76a514464b823812effd45
;path=/CPANInternal-140/DBIx-Class/t/resultset_overload.t
```

- ▶ `.t` is shared by Perl, Raku, Terra, Turing → Linguist rule `h/linguist/.t/0` matched → **Perl**
- ▶ The bytes were fetched at the pinned commit and their `sha1_git` verified; a git commit SHA **is** an SWH revision ID
- ▶ Verification of all samples via SWH `/known/`: content **261/261**, anchor commit **251/261**, origin **233/261**
- ▶ **Gap**: the "seen  $N \times$  in SWH" count comes from a filename + length match, not a strict content-id match; fixing it needs the nodes table (~840 GB) or an SWH-side index

# Appendix · the LLM contest protocol

```
turn: add Icon (+1 program)
```

```
List-Digest: 95f48d04
```

```
Model: meta-llama/llama-3.1-70b-instruct
```

```
Temperature: 0.4
```

```
turn: add Forge (+1 program)
```

```
List-Digest: d2538df5
```

```
Model: claude-sonnet-4-6
```

```
Agent: claude-code
```

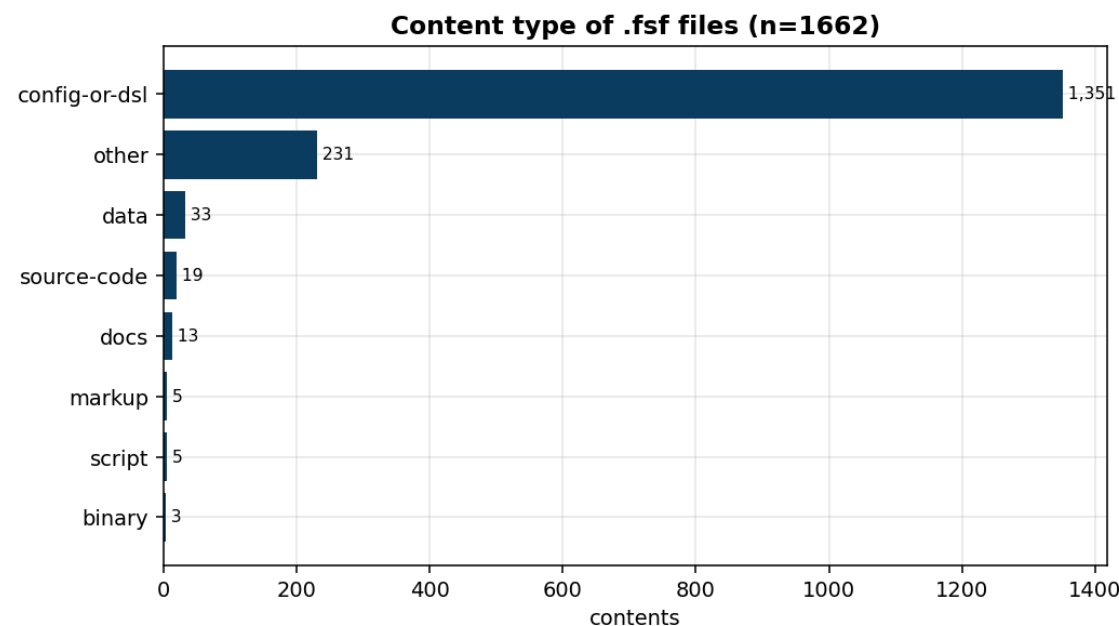
```
WebSearch: disabled
```

```
Strategy: batch-recall-100
```

- ▶ **List-Digest** pins the list the model saw; hooks reject duplicates (case-insensitive) and bad hashes
- ▶ 3,846 **turn: add** commits since Oct. 2025: Claude 3,392 · Grok 246 · Gemini 144 · Llama 48 · GPT/Codex 32 · DeepSeek 9 · Mistral 2
- ▶ Well-known languages are exhausted: agents now dig into regional languages, niche DSLs, historical systems

# Appendix · .fsf: an embedded DSL for neuroimaging

- ▶ **FSL FEAT** design files: fMRI analysis configuration
- ▶ a **declarative DSL** hosted in **Tcl** (`set fmri(...)` value)
- ▶ 1,662 files judged: **99%** configuration / DSL rather than general-purpose code; Tcl in 77%
- ▶ provenance: OpenNeuro, neuroimaging labs; top repo 18%
- ▶ polysemy tail: **GLSL fragment shaders**, git-annex / DataLad pointers
- ▶ a one-line marker (`set fmri()`) agrees with the judge **100%**



# Appendix · the review app

COBOL labels

browse

disagreements

unreviewed

with origin

dashboard

2783

labelled contents

2783

cached bytes

1

human-reviewed

2780

recovered origins

459

rule-labelled

is-COBOL label sources — Judge: the LLM verdict · reclassifier: deterministic content rules (reclassify.py) · oracle: LLM is-COBOL used to validate the reclassifier · division-gate: cheap baseline = has ≥2 COBOL divisions · human: your reviews. A file is Δ when its available is-COBOL votes disagree.

GROUP RULES

not-cobol:synthetic | origin = https://gitlab.com/tbetestpublic/repo-with-many-files-in-one-tree.git (459 contents) · WBC\_\*.FOO.CBL auto-generated placeholder (This is cobol file number N)

DATASETS

1k: 1000 | lo: 350 | main: 350 | pilot: 100 | eval:tail: 92 | eval:control: 50 | eval:wbc: 20

IS-COBOL AGREEMENT

Judge vs reclassifier: 2080 agree · 1 disagree

reclassifier vs LLM-oracle: 143 agree · 19 disagree

RECLASSIFIER LABEL DISTRIBUTION

|                       |      |  |
|-----------------------|------|--|
| cobol                 | 2080 |  |
| synthetic-placeholder | 459  |  |
| other                 | 122  |  |
| comic-book-list       | 45   |  |
| cobol-copybook        | 28   |  |
| binary-data           | 25   |  |
| cobol-generated       | 24   |  |

One pane over every label for a file: LLM judge · reclassifier · gate · human · origin. Disagreements are surfaced; group rules label a whole origin at once.

.rpgle labels

browse

non-RPGLE

unreviewed

rule-labelled

/SAX2.rpgle

/SAX2.rpgle

swh:1:cnt:f687e99428d1d0bb8329ea4b129289f6921f05a · 13603 bytes · SWH\_

\* Summary: SAX2 parser interface used to build the DOM tree

\* Description: those are the default SAX2 interfaces used by the library when building DOM tree.

\*

\* Copy: See Copyright for the status of this software.

\*

\* Author: Patrick Monnerat <pm@datasphere.ch>, DATASPHERE S.A.

/if not defined(XML\_SAX2\_H\_)

/define XML\_SAX2\_H\_

/include "libxmlrpg/xmlversion"

/include "libxmlrpg/parser"

/include "libxmlrpg/xlink"

d xmlSAX2GetPublicId...

d pr

d ctx

d value

d extproc('xmlSAX2getPublicId')

d const xmlChar \*

d void \*

d xmlSAX2GetSystemId...

d pr

d ctx

d value

d extproc('xmlSAX2getSystemId')

d const xmlChar \*

d void \*

d xmlSAX2SetDocumentLocator...

d pr

d ctx

d value

d extproc('xmlSAX2SetDocumentLocator')

d void \*

d value like(xmlSAXLocatorPtr)

d xmlSAX2GetLineNumber...

d pr

d ctx

d value

d extproc('xmlSAX2GetLineNumber')

d void \*

d value

d xmlSAX2GetColumnNumber...

d pr

d ctx

d value

d extproc('xmlSAX2GetColumnNumber')

d void \*

d value

d xmlSAX2IsStandalone...

d pr

d ctx

d value

d extproc('xmlSAX2IsStandalone')

d void \*

d value

d xmlSAX2HasInternalSubset...

d pr

d ctx

d value

d extproc('xmlSAX2HasInternalSubset')

d void \*

d value

ORIGIN

https://github.com/kiyolee/libxml2 | github.com

LLM JUDGE

content\_type

language

source\_format

unit\_kind

related\_languages

embedded SQL

platform

domain / maturity

is PL?

copybook-or-header

RPG IV / ILE RPG (fixed-format)

fixed-format

copybook-prototype-header

RPG, C

False

ibm-i-as400

library-framework · library-quality

True

Prototype header (copybook) for the libxml2 SAX2 parser interface, providing RPG IV procedure prototypes (PR definitions with EXTPROC) that bind to the C-language xmlSAX2\* functions in libxml2. Intended to be /include-d by RPG programs that need to call the SAX2 DOM-building callbacks. Uses conditional compilation directives to conditionally expose SAX1-compatible and HTML/DOCB handler prototypes.

INDICATORS

lines

comment ratio

\*\*FREE

format guess

fixed spec lines

/free blocks

dol- / dol-proc / dol-pr

EXEC SQL

/copy /include

looks copybook

reclass

248

0.034

False

fixed-format

175

0

0 / 0 / 0

False

3

True

rpgle-copybook

ASSERT FOR A GROUP

One statement labels every matching content (e.g. all files from a non-RPGLRE repo).

Scope

this origin

Value

https://github.com/kiyolee

Label

not-rpgle

Buttons

```
{
  "schema": 1,
  "subject": { "sha1_git": "f851a314...", "filename": "XSetModifierMapping.m", "ext": ".m" },
  "reviewer": { "kind": "human", "id": "mathieu-acher" },
  "verdict": { "label": "pl/new:tet-scenario", "confidence": "high", "supersedes": null },
  "comment": "xts5 TET 'mc' format: C embedded in >>CODE blocks.",
  "shown": { "claimants": ["pl/m4", "pl/matlab", "pl/mercury", "..."] },
  "created_at": "2026-06-12T09:31:02Z"
}
```

- ▶ `reviews/<sha1_git>/<stamp>--<reviewer>--<hash>.json`: one immutable file per review, never a merge conflict
- ▶ reviewers are **humans, tools or LLMs**; what the UI showed is recorded (anchoring-bias trace)
- ▶ consensus is **derived**, and can be recomputed under other rules